

NAKED-EYE ASTRONOMY

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OPTICAL INSTRUMENTS ARE NOT NECESSARY to gain a foothold in astronomy—our ancestors did without them for thousands of years. Today's naked-eye observer, equipped with a little foreknowledge and some basic equipment, can still appreciate the constellations, observe the brightest deep-sky objects, and trace the paths of the Moon and planets in the night sky.

PREPARING TO STARGAZE

To get the most from stargazing, some preparation is needed. The human eye takes some 20 minutes to adjust to darkness and, as the pupil opens, more detail and fainter objects become visible. Look at a planisphere or monthly sky chart (see pp.426–501) to see what is currently in the sky. A good location is one shielded from street lights and ideally away from their indirect glow. Try to avoid all artificial light—if necessary, use a flashlight with a red filter. Keep a notebook or a prepared report form to record observations, especially if looking for particular phenomena, such as meteors. To see faint stars and deep-sky objects, avoid nights when a bright Moon washes out the sky. Even on a dark, cloudless night, air turbulence can affect the observing quality or “seeing”—the

SEEING AND TWINKLE

Variable “seeing” is caused by warm air currents rising from the ground at nightfall. These telescope images of Jupiter show the range of seeing from poor to fine, but seeing also limits the visibility of stars with the naked eye and determines the amount of “twinkle.”



best nights are often those that do not suddenly get colder at sunset.

LIGHT POLLUTION

This composite satellite image shows the extent of artificial lighting on Earth. In industrialized regions, it is almost impossible to find truly dark skies.



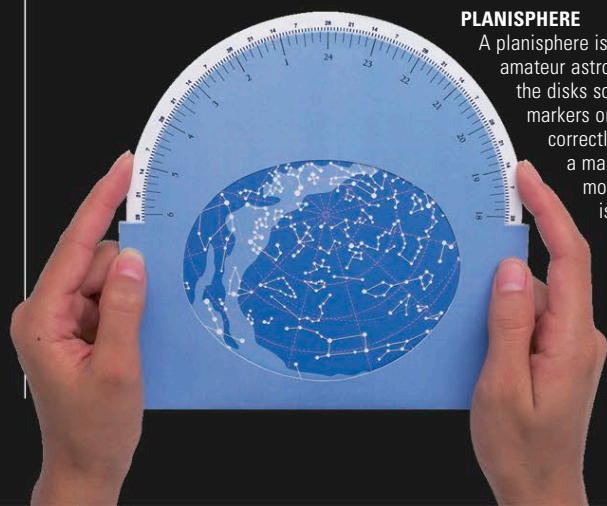
GOOD STREET LIGHTING

In some countries, non-essential street lights are switched off late at night. Elsewhere, shades are installed to project all the light downward, preventing it from leaking into the sky. Such measures can increase the light on the street, save energy, and preserve the night sky for stargazers.



PLANISPHERE

A planisphere is a useful tool for any amateur astronomer. The user rotates the disks so that the time and date markers on the edge match up correctly, and the window reveals a map of the sky at that moment. A single planisphere is useful only for a limited range of latitudes, so be sure to get one with the correct settings.



THE MOON AND VENUS

Solar system objects such as the Moon and Venus can be spectacular sights even with the unaided eye. This beautiful twilight pairing was photographed in January 2004.